Hukum Biot-Savart

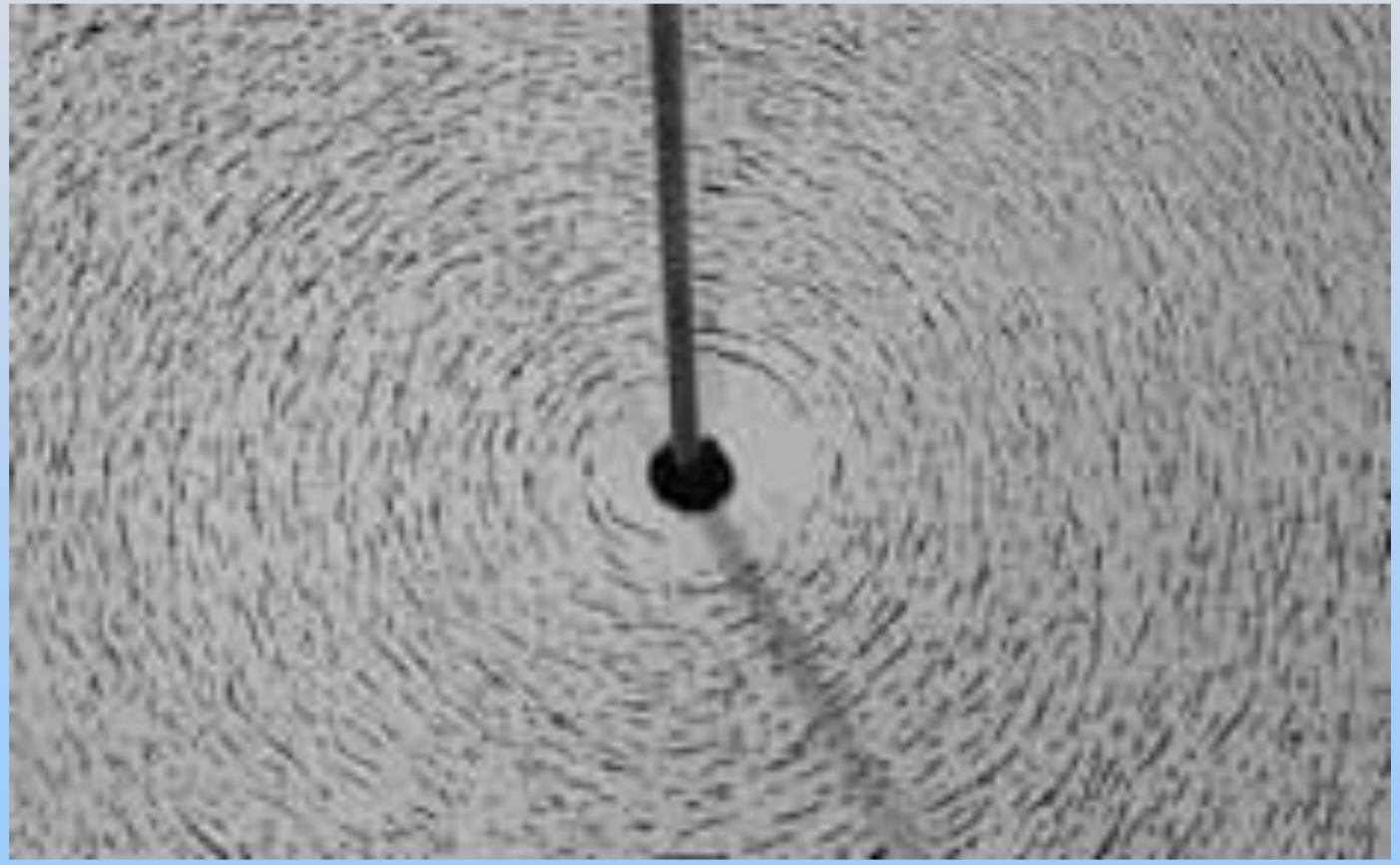
Dosen : Defrianto Pratama, S.Pd, M.Si.

Teknik Elektronika

Politeknik Negeri Bandung

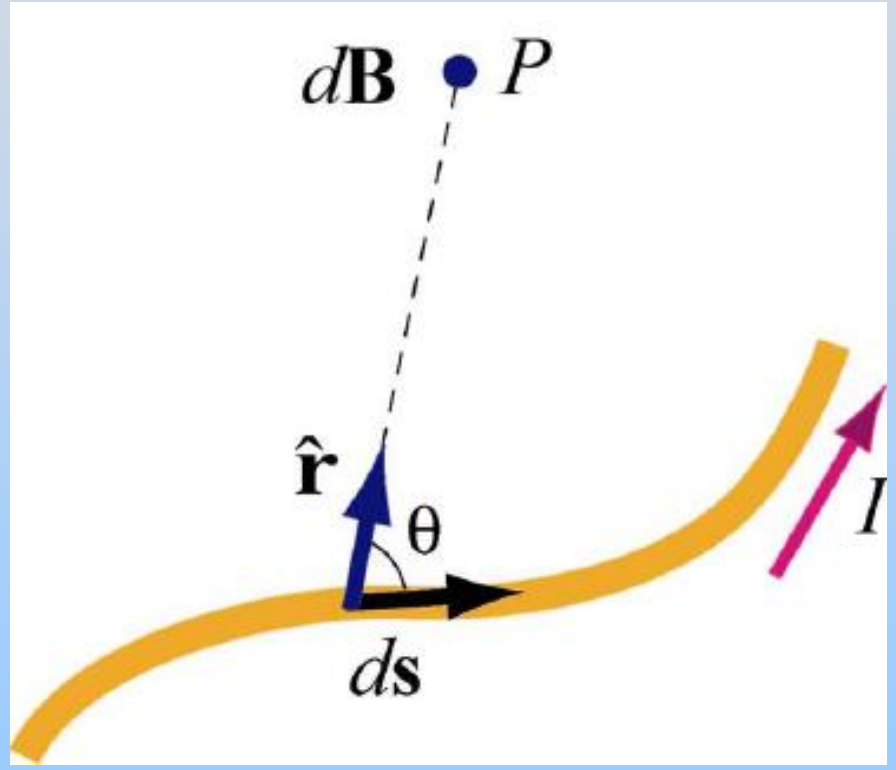
Sumber bacaan Ebook: Halliday ch 29

Medan Magnet di Sekitar Kawat Berarus



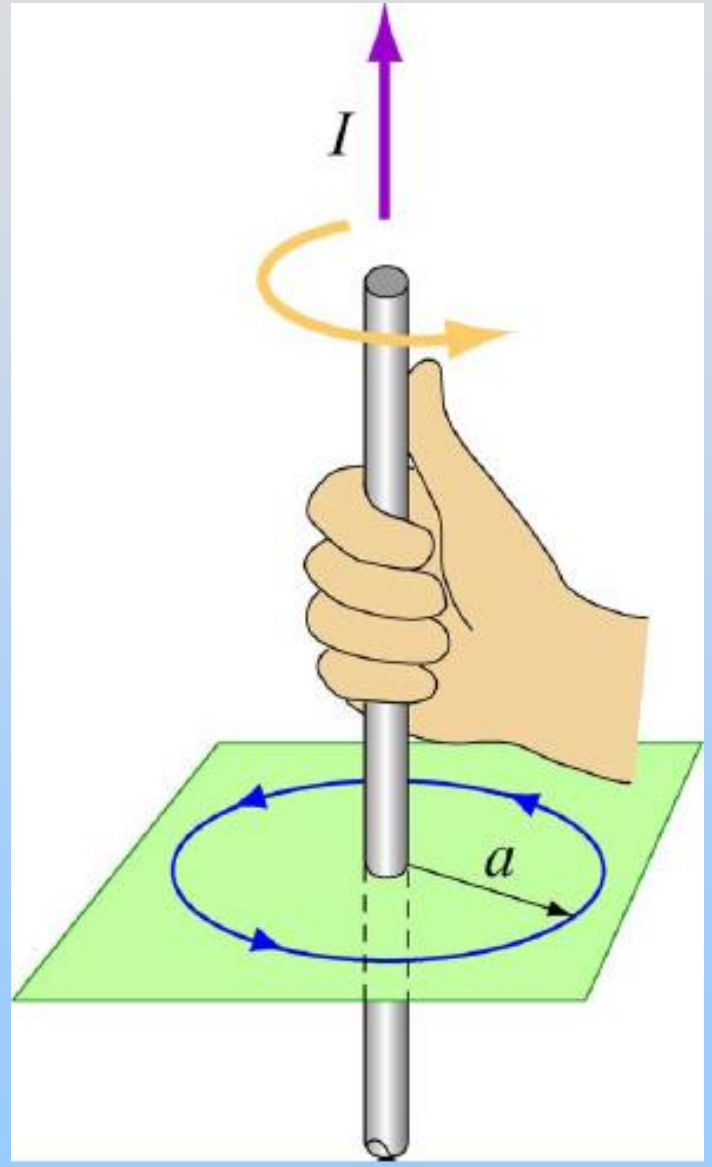
Hukum Biot-Savart

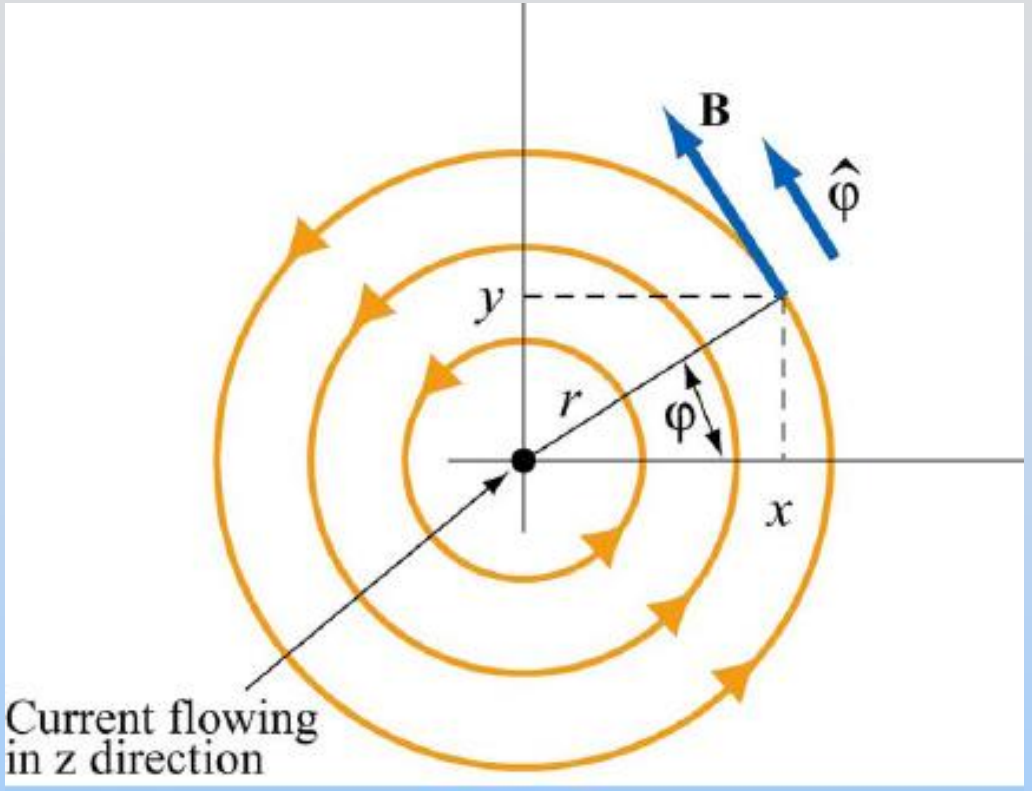
Elemen arus sepanjang ds yang membawa arus I menghasilkan medan magnet



$$d\vec{\mathbf{B}} = \frac{\mu_0}{4\pi} \frac{I\, d\vec{\mathbf{s}} \times \hat{\mathbf{r}}}{r^2}$$

Aturan Tangan Kanan 2



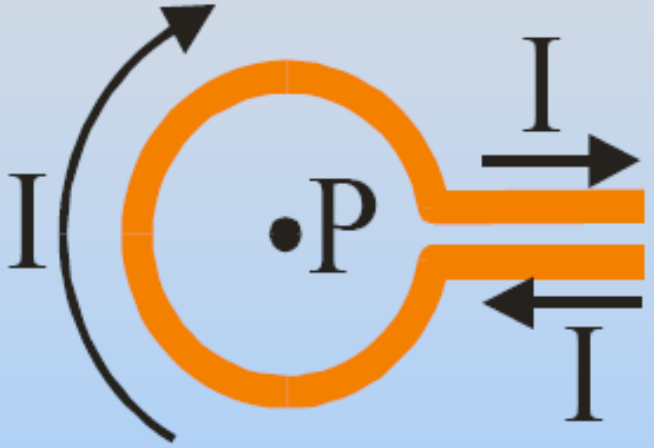


$$\hat{\mathbf{z}} \times \hat{\boldsymbol{\rho}} = \hat{\boldsymbol{\varphi}}$$

Animasi 7.7

Contoh : Koil Radius *R*

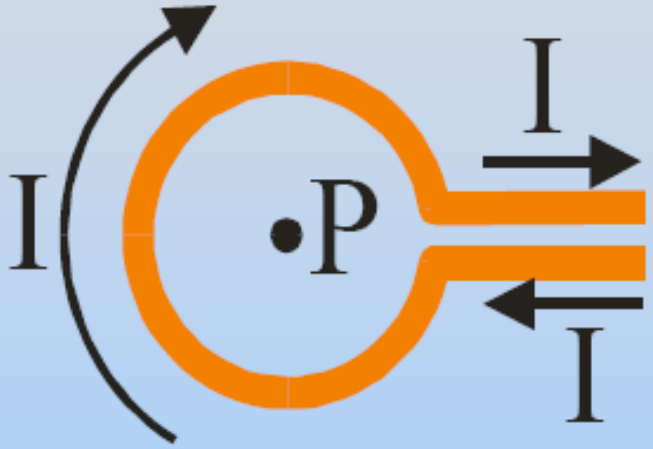
Tinjau sebuah koil dengan radius *R* dan arus *I*



Carilah medan magnet **B** di pusat koil (P)!

Contoh : Koil Radius *R*

Tinjau sebuah koil dengan radius *R* dan arus *I*

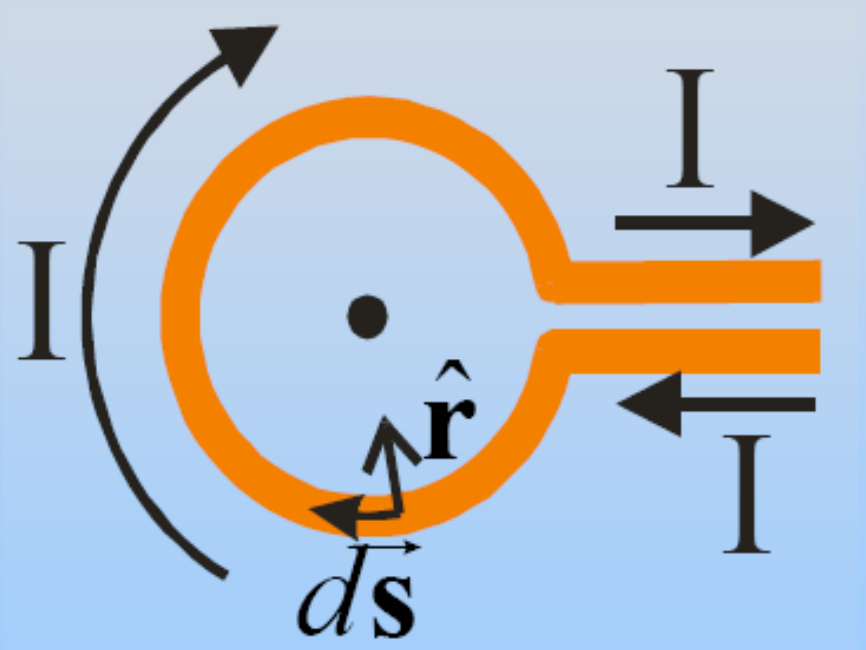


1) Pikirkan sejenak “arahnya”
2) Pilih ds
3) Tetapkan sistem koordinat
4) Tulis hukum Biot-Savart

Contoh : Koil Radius *R*

Bagian yang melingkar pada koil…

$$d\vec{\mathbf{s}} \perp \hat{\mathbf{r}} \quad \rightarrow \quad \left| d\vec{\mathbf{s}} \times \hat{\mathbf{r}} \right| = ds$$



Biot-Savart:

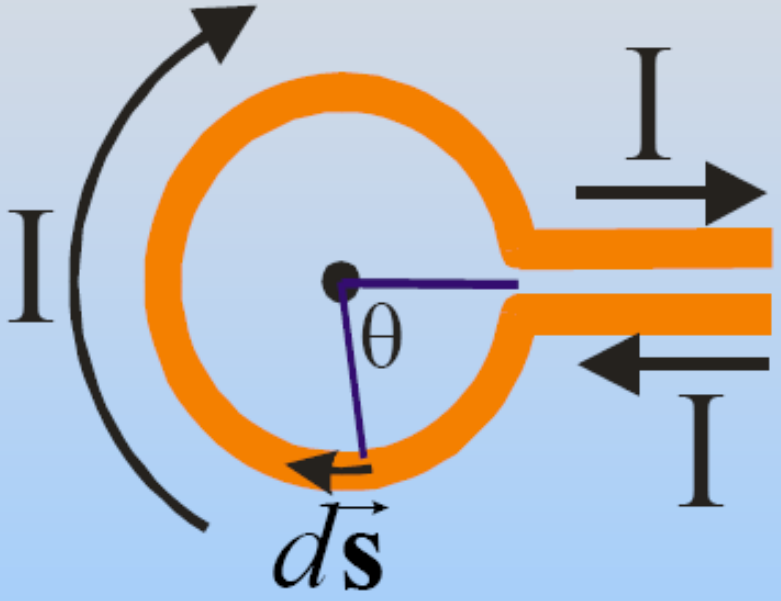
$$dB = \frac{\mu_0 I}{4\pi} \frac{\left| d\vec{\mathbf{s}} \times \hat{\mathbf{r}} \right|}{r^2} = \frac{\mu_0 I}{4\pi} \frac{ds}{r^2}$$

$$= \frac{\mu_0 I}{4\pi} \frac{R\, d\theta}{R^2}$$

$$= \frac{\mu_0 I}{4\pi} \frac{d\theta}{R}$$

Contoh : Koil Radius *R*

Tinjau sebuah koil dengan radius *R* dan arus *I*



$$dB = \frac{\mu_0 I}{4\pi}\frac{d\theta}{R}$$

$$B = \int dB = \int_0^{2\pi} \frac{\mu_0 I}{4\pi}\frac{d\theta}{R}$$

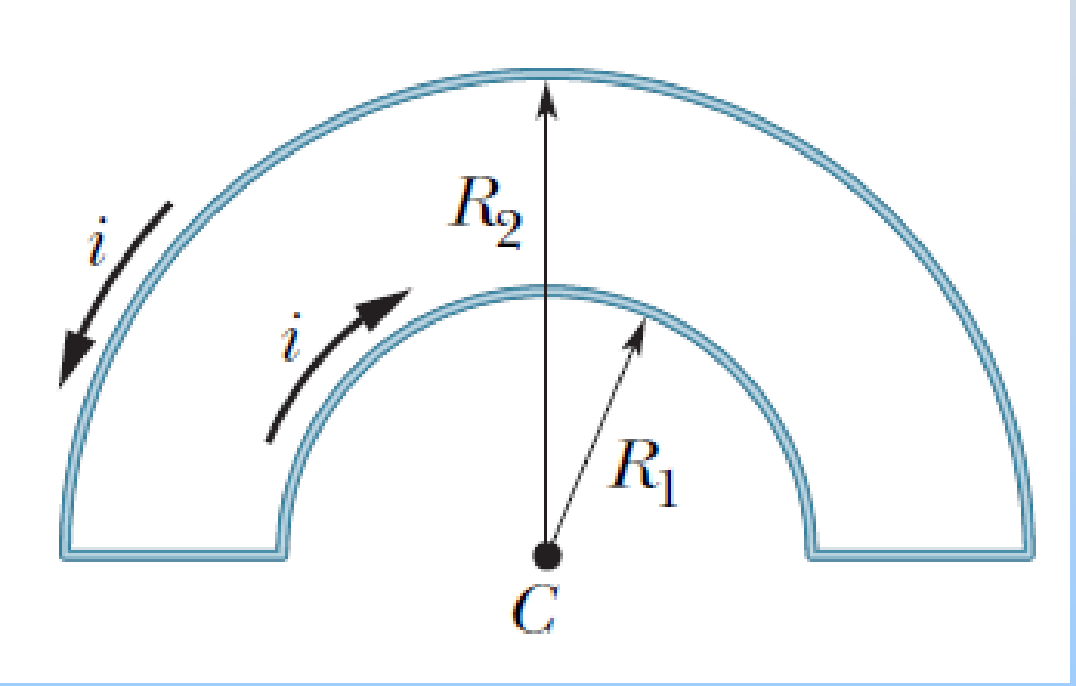
$$= \frac{\mu_0 I}{4\pi R}\int_0^{2\pi} d\theta = \frac{\mu_0 I}{4\pi R}(2\pi)$$

$$\vec{\mathbf{B}} = \frac{\mu_0 I}{2R}$$

Masuk bidang

TUGAS NO 1

- Tentukan besar medan magnet B pada titik c nyatakan dalam arus i, R1 dan R2



TUGAS NO 2

Medan B dari Koil Radius R

